

C227

Research Proposals

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A doctoral student who presents at a regional conference each spring is turning her approved proposal into her final research report. Which content must she add that the proposal did not contain?

- A. A methods section
- B. Nothing; the report is always identical to the proposal.
- C. Results and a discussion of the findings
- D. A literature review

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A reader wants the part of a finished report that corresponds to the proposal's "Proposed Methods." Where should she look?

- A. Nowhere, because methods are removed once data are collected.
- B. The methods section, which now describes what was actually done.
- C. The results section
- D. The discussion section

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

In a proposal for a committee that includes a statistician, a student writes, "Surveys were given to 200 parents, and responses were analyzed with regression." What is the problem with this sentence?

- A. It should be in present tense, because proposals describe current work.
- B. Nothing; proposals and reports always use the same tense.
- C. Regression cannot be named in a proposal.
- D. It is in past tense; a proposal should say surveys will be given and responses will be analyzed.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

The study is now finished. Which sentence belongs in the methods section of the final report?

- A.** Surveys are given to 200 parents in March.
- B.** Surveys will be given to 200 parents in March.
- C.** Surveys were given to 200 parents in March.
- D.** Surveys should be given to 200 parents in March.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

At a school that just opened a new STEM wing, a proposal's problem is low math engagement among English learners. Its research question is, "How do English learners describe their engagement in math class?" The planned design is a randomized experiment measuring test scores. What is the main flaw?

- A.** No flaw; experiments can answer any question.
- B.** The design does not match the question; describing experience calls for interviews or observation.
- C.** The problem should be rewritten around test scores to fit the experiment.
- D.** It needs a longer literature review.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A committee member at a university with a new education building asks a student what her methods section should include. Which list is correct?

- A.** Participants and sampling, instruments, procedures, and the data analysis plan
- B.** Background and a summary of prior studies
- C.** Only the survey instrument
- D.** Expected findings and their implications

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A researcher who taught high school English for 12 years studies teacher burnout. She chooses to include only teachers in their first five years who work in one state. How should this be labeled in the proposal?

- A. Nothing; design choices never need to be reported.
- B. A limitation
- C. An assumption
- D. A delimitation

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

Which statement describes a limitation rather than a delimitation?

- A. Participants are assumed to answer honestly.
- B. The study includes only first-year teachers.
- C. The study focuses on a single state.
- D. Only 38% of invited teachers returned the survey.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A researcher plans to survey tenth graders about social media use at a high school with a nationally ranked debate team. Before collecting any data, what is required?

- A. IRB approval, parent or guardian permission, and student assent
- B. Student consent alone, since teenagers can decide for themselves
- C. The principal's approval in place of IRB review
- D. Nothing, because surveys are always exempt from review

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A researcher collected survey data in October and submitted the study to the IRB in November. What is the problem?

- A. The IRB reviews only final reports, so timing does not matter.
- B. No problem, as long as approval is granted later.
- C. Data were collected before approval, so they generally cannot be used.
- D. It matters only if a participant complains.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A student who works as an instructional coach is drafting the Significance of the Study section of her proposal. Which sentence belongs there?

- A.** Findings may help district coaches decide how to structure feedback cycles for new teachers.
- B.** This study will definitely transform teacher retention nationwide.
- C.** The study found that feedback cycles raised retention by 20%.
- D.** Prior studies found that feedback cycles vary widely across districts.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A proposal studies “student engagement” at a middle school that recently won a state award. Which is an operational definition?

- A.** Being really into learning.
- B.** The percentage of 30-second observation intervals in which a student is on task, recorded on a structured form.
- C.** A dictionary definition of engagement.
- D.** Engagement will be obvious to any observer.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A doctoral student ~~who presents at a regional conference each spring~~ is turning her approved proposal into her final research report. Which content must she add that the proposal did not contain?

Target word: what the report adds **Decides:** Only the report has data to present and interpret.

A	Confusion set	Both contain methods; only the tense changes.
B	Absolute language	The report adds results and discussion.
C	Correct	The proposal cannot report results that do not exist yet.
D	Adjacent category	Both documents contain one.

2. Answer B Concept 1 • transfer

A reader wants the part of a finished report that corresponds to the proposal's "Proposed Methods." Where should she look?

Target word: reverse mapping **Decides:** Methods carry into the report in past tense.

A	Dropped element	Methods stay so others can judge and repeat the study.
B	Correct	Methods persist, updated to the completed procedures.
C	Adjacent category	Results report what the data showed, not how they were gathered.
D	Confusion set	Discussion interprets findings.

3. Answer D Concept 2

In a proposal ~~for a committee that includes a statistician~~, a student writes, "Surveys were given to 200 parents, and responses were analyzed with regression." What is the problem with this sentence?

Target word: tense in a proposal **Decides:** The study has not happened yet.

A	Confusion set	The work is planned, so future tense fits.
B	Absolute language	The report shifts to past tense after the study.
C	Adjacent category	Naming the planned analysis is required.
D	Correct	Proposals describe planned work in future tense.

4. Answer C Concept 2 • transfer

The study is now finished. Which sentence belongs in the methods section of the final report?

Target word: reverse: tense after the study **Decides:** Completed work is reported in past tense.

A	Confusion set	Present tense suggests ongoing work.
B	Inverted operation	Future tense belongs to the proposal.
C	Correct	The report describes what was done, in past tense.
D	Adjacent category	A recommendation, not a description of method.

5. Answer B Concept 3

~~At a school that just opened a new STEM wing,~~ a proposal's problem is low math engagement among English learners. Its research question is, "How do English learners describe their engagement in math class?" The planned design is a randomized experiment measuring test scores. What is the main flaw?

Target word: alignment **Decides:** An experience question needs a qualitative design.

A	Absolute language	Experiments test effects, not lived experience.
B	Correct	Question type should decide the method.
C	Inverted operation	Bends the problem to fit the method instead of the reverse.
D	Adjacent category	Length is not the problem; alignment is.

6. Answer A Concept 4

A committee member ~~at a university with a new education building~~ asks a student what her methods section should include. Which list is correct?

Target word: methods contents **Decides:** Who, with what, how, and how analyzed.

A	Correct	These let readers judge and repeat the study.
B	Confusion set	Those belong in the introduction and literature review.
C	Dropped element	Drops participants, procedures, and analysis.
D	Inverted operation	Findings come after data, not in the plan.

7. Answer D Concept 5

A researcher ~~who taught high school English for 12 years~~ studies teacher burnout. She chooses to include only teachers in their first five years who work in one state. How should this be labeled in the proposal?

Target word: classify a chosen boundary **Decides:** She chose it.

A	Absolute language	Scope choices must be stated so readers know the boundaries.
B	Confusion set	Limitations are outside the researcher's control.
C	Adjacent category	Assumptions are things taken as true, not scope choices.
D	Correct	A boundary set by the researcher.

8. Answer D Concept 5 • transfer

Which statement describes a limitation rather than a delimitation?

Target word: reverse: find the uncontrolled weakness **Decides:** Something the researcher could not control.

A	Adjacent category	An assumption.
B	Inverted operation	A chosen boundary, a delimitation.
C	Confusion set	Also a chosen boundary.
D	Correct	Response rate is outside the researcher's control.

9. Answer A Concept 6

A researcher plans to survey tenth graders about social media use at a high school ~~with a nationally ranked debate team~~. Before collecting any data, what is required?

Target word: ethics requirements with minors **Decides:** Participants are minors.

A	Correct	Minors need both parental permission and their own agreement, after IRB approval.
B	Dropped element	Drops parental permission for minors.
C	Adjacent category	Site permission does not replace ethics review.
D	Absolute language	Exemption is decided by the IRB, not assumed.

10. Answer C Concept 6 • transfer

A researcher collected survey data in October and submitted the study to the IRB in November. What is the problem?

Target word: order of approval and data collection **Decides:** Approval must come first.

A	Adjacent category	The IRB reviews planned studies before they start.
B	Inverted operation	Reverses the required order.
C	Correct	IRB review must precede data collection.
D	Confusion set	The rule applies regardless of complaints.

11. Answer A Concept 7

A student ~~who works as an instructional coach~~ is drafting the Significance of the Study section of her proposal. Which sentence belongs there?

Target word: significance content **Decides:** States potential value without claiming results.

A	Correct	Names who could benefit and why it matters.
B	Absolute language	Overclaims what one study can do.
C	Inverted operation	Reports results that do not exist yet.
D	Confusion set	That belongs in the literature review.

12. Answer B Concept 8

A proposal studies “student engagement” at a middle school ~~that recently won a state award~~. Which is an operational definition?

Target word: operational definition **Decides:** States exactly how the term will be measured.

A	Dropped element	Drops any way to measure it.
B	Correct	Specifies the measurement procedure.
C	Adjacent category	A conceptual definition, not an operational one.
D	Absolute language	Without a procedure, observers will disagree.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	10	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	7	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Proposal sections map to report sections	Introduction, literature review, and methods carry into the report; the report adds results and discussion, which a proposal cannot contain.	Thinking the report simply repeats the proposal, or that methods disappear.	Converting an approved proposal into a final report.
2	Proposal tense	A proposal describes planned work in future tense (“will be surveyed”); a report describes completed work in past tense.	Writing the proposal as if the study already happened.	A method sentence in a proposal draft.
3	Alignment	Problem, purpose, research questions, design, and analysis must fit one another; the question type decides the method.	Pairing an experience question with an experiment, or bending the problem to fit a favorite method.	A proposal whose question and method point in different directions.
4	What the methods section contains	Methods state participants and sampling, instruments, procedures, and the data analysis plan.	Putting background or expected findings in methods.	“What belongs in the methods section?”
5	Limitations vs. delimitations	Delimitations are boundaries the researcher chooses; limitations are weaknesses outside the researcher’s control.	Calling a chosen boundary a limitation.	“Chooses to include only ...” vs. “only 38% responded.”
6	Ethical approval and consent	IRB approval comes before any data collection; minors need parent or guardian permission plus their own assent.	Collecting data first or treating school permission as a substitute for IRB review.	A study with minors or a timeline of approval and data collection.
7	Significance of the study	The significance section explains who could benefit and why the question matters, without claiming results.	Reporting findings or reviewing literature in the significance section.	“Which sentence belongs in the significance section?”
8	Operational definitions	An operational definition states exactly how a construct will be observed or measured in this study.	Offering a dictionary or everyday definition.	A key term like engagement or motivation that must be measured.

Course scope (public description): Research Proposals focuses on planning and writing a well-organized and complete research proposal. The relationship of the sections in a research proposal to the sections in a research report will be highlighted. Research Design and Analysis is a prerequisite for this course.